

Redox Flow Batteries

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High-Power Near-Neutral Aqueous All Organic Redox Flow Battery Enabled with a Pair of Anionic Redox Species

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Abstract: Aqueous organic redox flow batteries (AORFBs) are regarded as a promising alternative for low-cost and durable grid-scale energy storage. However, the narrow potential gap, chemical lability and membrane fouling in most AORFBs constitute formidable roadblocks for practical applications. Herein, a pair of anionic organic molecules, namely (PPBPY)Br₂ and PSS-TEMPO, are proposed. The (PPBPY)Br₂ in anolyte reveals remarkable electrochemical stability without degradation after 1000 cycles, while PSS-TEMPO in catholyte presents a capacity decay rate as low as 0.012 %/cycle. At near-neutral conditions, the (PPBPY)Br₂/PSS-TEMPO flow cell exhibits a high voltage of 1.61 V, extremely low permeability across cation-exchange membrane and thus excellent cycling stability. Notably, a highest peak power density of 509 mWcm⁻² has been achieved among reported all-organic aqueous RFBs. The molecular engineering strategies demonstrated here could provide a credible example of high-performance AORFBs.

Introduction

“Carbon neutrality” has been in dire need to address global warming and the consequent environmental and societal challenges, which is thus being pledged by many countries to reduce carbon emissions. As a result, the penetration of renewable energy sources, such as solar and wind, has rapidly escalated in recent years than ever before. However, high levels of power generation from these non-dispatchable energy sources can be challenging when directly integrated into electric power grids, which may impose a large impact on the grids or result in curtailment. Therefore, advanced electrical energy storage systems are urgently needed to buffer the temporal deviation in renewable energy production and consumption.^[1] Various battery technologies have been employed for electricity storage. With the silent features of superior scalability, operation flexibility (decoupled energy storage and power generation) and safety, redox flow battery (RFB) is regarded as a promising

candidate for large-scale energy storage applications.^[2] Among various RFB technologies, all-vanadium flow battery (VFB), as the most developed one, has been demonstrated in megawatt scales.^[3] However, the limited vanadium resources which render high materials cost, along with other hurdles such as the use of corrosive electrolyte and poor thermal stability, impede the widespread applications of VFB.

In recent years, aqueous organic RFBs (AORFBs), due to the tunable properties of organic redox species (*i.e.*, potential, solubility, permeability, *etc.*) and potentially low cost, have been demonstrated as a promising alternative to address the critical issues of VFB.^[4] Among various reported AORFBs, 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) and 1,1'-disubstituted-4,4'-bipyridilium (viologen) derivatives have attracted considerable attention.^[5] The first viologen/TEMPO-based flow battery used 4-hydroxy-TEMPO in catholyte and methyl viologen (MV) in anolyte separated by an anion-exchange membrane (AEM), which revealed high cell voltage, fast reaction kinetics, and good solubility in pH-neutral conditions.^[6] However, the MV/4-hydroxy-TEMPO pair encountered problems of redox species crossover through AEM and bimolecular decomposition of MV. Since then, several modifications have been applied to these two molecules (See Table 1). Ammonium-based groups are mostly used in viologen derivatives and polymerized ones,^[7] with which the molecules become positively charged with increased repulsion among the redox species and with AEMs, consequently showing enhanced stability and reduced crossover. Similar strategies have also been applied to TEMPO to prepare positively charged

Table 1: TEMPO and viologen derivatives reported in literature and studied here for AORFBs.

Redox molecule	Oxidized/reduced state	Redox potential (V vs. SHE)	Membrane
TEMPO derivatives			
TEMPTMA ^[13]	+2/+1	0.790	AEM
TMAP-TEMPO ^[14]	+2/+1	0.810	AEM
Pyr-TEMPO ^[15]	+2/+1	0.810	AEM
4-CO ₂ Na-TEMPO ^[12a]	0/-1	0.790	CEM
PSS-TEMPO	0/-1	0.805	CEM
Viologen derivatives			
(APBPY)Cl ₄ ^[16]	+4/+2	-0.763	AEM
(PyrPV)Cl ₄ ^[15]	+4/+2	-0.760	AEM
BTMAP-Vi ^[17]	+4/+3	-0.358	AEM
(SPR) ₂ V ^[9]	0/-1	-0.430	CEM
(PPBPY)Br ₂	-2/-3	-0.801	CEM

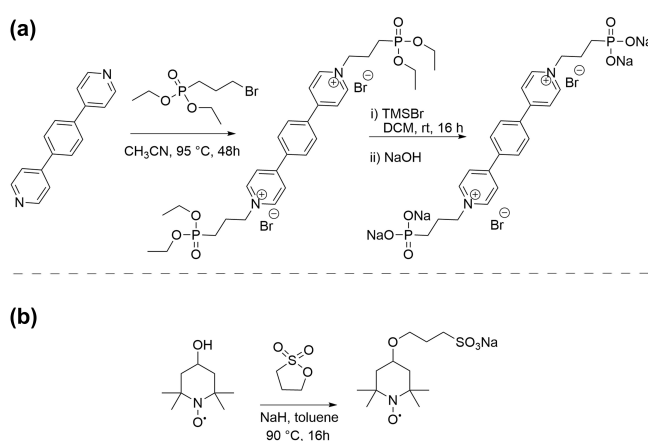
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derivatives. However, for these positively charged redox molecules, the use of AEMs leads to reduced power density due to relatively high membrane resistance, while the cation-exchange membranes (CEMs) would cause membrane fouling.^[8] In 2018, Liu group reported 1,1'-bis(3-sulfonatopropyl)-4,4'-bipyridinium ((SPr)₂V), which was compatible with CEMs.^[9] Nevertheless, because of the low charge in the oxidized (0) and reduced state (−1), (SPr)₂V still suffers from a relatively fast crossover through the CEM. In 2020, Aziz group introduced a phosphonated viologen derivative, 1,1'-bis(3-phosphonopropyl)-[4,4'-bipyridine]-1,1'-dium dibromide to address the above problem,^[10] but only an areal power of 143 mW cm^{−2} was achieved partly due to the low cell voltage. Note that the similar strategy has also been implemented with ferrocene derivatives, while these sulfonated anionic species generally have relatively low redox potentials when used in catholyte, thus also leading to a low cell voltage.^[11]

Herein, we report a phosphonate functionalized phenylene-bridged viologen derivative 1,1'-bis(3-phosphonopropyl)-4,4'-(1,4-phenylene) bipyridinium dibromide ((PPBPy)Br₂) as the anolyte redox active species. It has a redox potential of −0.801 V vs. SHE and extremely low permeability through CEMs. (PPBPy)Br₂ has also demonstrated remarkably stable cycling performance, for which the half-cell test literally showed no capacity decay after 1000 cycles and undetectable structure changes. On the cathode side, 4-(propoxy-3-sodium sulfonate)-2,2,6,6-tetramethyl piperidin-1-yl)oxyl (PSS-TEMPO) was synthesized. Compared with the reported 4-CO₂Na-TEMPO and 4-SO₃K-TEMPO,^[12] the longer side chain of PSS-TEMPO effectively alleviates the adverse impact of the strong electron-withdrawing group on the skeleton stability, and provides effective steric hindrance suppressing the interactions between TEMPO radicals, also ensuring a lower permeability through CEMs. An AORFB system compatible with CEMs at near-neutral conditions has been demonstrated by pairing (PPBPy)Br₂ and PSS-TEMPO, which presents a cell voltage as high as 1.61 V and the highest power density of 509 mW cm^{−2} among all the reported pH-neutral aqueous RFBs. Meanwhile, this AORFB has also exhibited an outstanding cycling performance with barely species crossover. We anticipate this work provides a promising and credible example for high-performance AORFBs for stationary energy storage.

Results and Discussion

The synthetic approaches of (PPBPy)Br₂ and PSS-TEMPO are shown in Scheme 1. A quick and facile chemical synthesis enables easy access to the desired molecules for research and commercial application purposes. Therefore, we adopted no purification multiple-step and single-step synthetic methods for (PPBPy)Br₂ and PSS-TEMPO, respectively. (PPBPy)Br₂ was synthesized in two simple synthetic steps without resource-intensive chromatography requirement. Briefly, 1,4-di-(pyridin-4-yl)benzene (PhPy) was reacted with diethyl(3-bromopropyl)phosphonate in



Scheme 1. Synthetic routes for a) (PPBPy)Br₂ and b) PSS-TEMPO.

acetonitrile at 95 °C for 48 h, followed by a simple hydrolysis of the ester resulting in the desired product in 77 % overall yield. The PhPy precursor can be synthesized economically by a single-step Suzuki–Miyaura reaction of 1,4-dibromobenzene and 4-pyridineboronic acid pinacol ester with a very high yield (>90 %).^[18] PSS-TEMPO was made in a single step by using the commercially available 4-hydroxy TEMPO. 4-hydroxy TEMPO was reacted with 1,3-propanesultone in the presence of sodium hydride in toluene at 95 °C for 16 hours with a yield of 70 %. The structures of (PPBPy)Br₂ and PSS-TEMPO were characterized by Nuclear Magnetic Resonance (¹H NMR) and Mass Spectrometry (MS) analysis (See Figure S1–S4). Figure 1a,b depict the configuration of the AORFB configuration and cyclic voltammograms (CVs) of the two molecules. The redox potentials of (PPBPy)Br₂ and PSS-TEMPO are −0.801 V and 0.805 V vs. SHE respectively, offering a cell voltage of 1.606 V, which is considerably higher than the commonly reported AORFBs. Furthermore, (PPBPy)Br₂ has a solubility of above 1.10 M in water and that of PSS-TEMPO is above 2.55 M, corresponding to a volumetric capacity of 29.48 Ah L^{−1} and 68.43 Ah L^{−1} for anolyte and catholyte, respectively. Thus, the theoretical energy density of the (PPBPy)Br₂/PSS-TEMPO AORFB can achieve 33.1 Wh L^{−1}. Previous reports have suggested that the two-electron reductions of phenylene-bridged bipyridinium derivatives are reversible in aqueous solution.^[15,16] However, presumably due to the low redox potential (Figure S5), when (PPBPy)Br₂ is reduced to the second electron, severe hydrogen evolution reaction (HER) occurs. Thus, only the first reduction of (PPBPy)Br₂ was used in this work.

The diffusion coefficient (*D*) and electron transfer rate constant (*k*₀) of (PPBPy)Br₂ and PSS-TEMPO were investigated by both Koutecký–Levich approach and Nicholson method to ensure the accuracy of the results. Figure 2a,b show the CV curves of the two molecules at different scan rates, and Figure 2c,d show the linear sweep voltammetry (LSV) curves obtained with a rotating disk electrode (RDE). Note that the LSV scan potential of (PPBPy)Br₂ was cut at −0.940 V to avoid reaching the second reduction and HER. Based on the Nicholson method, Figure S6a–d

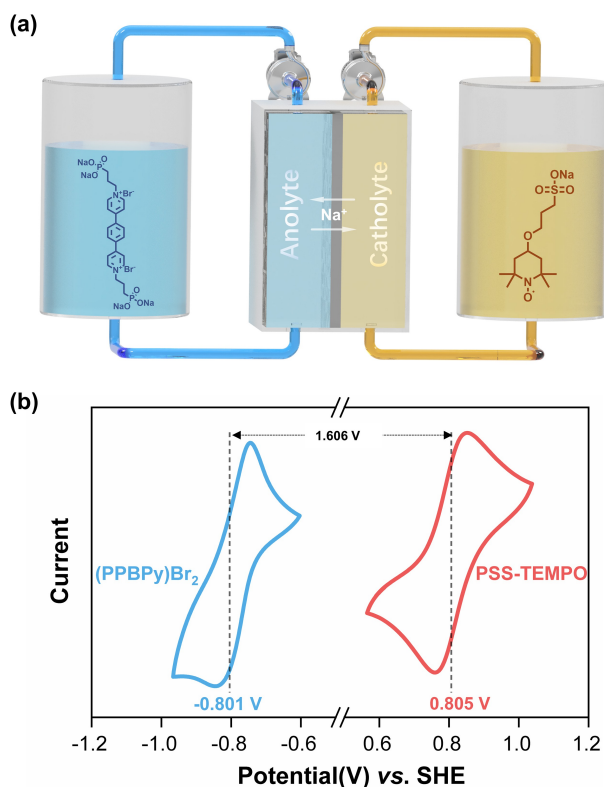


Figure 1. a) Schematic of the flow battery assembled with (PPBPy)Br₂ as anolyte and PSS-TEMPO as catholyte, with Na⁺ as charge carrier. b) CV curves of 20 mM (PPBPy)Br₂ (blue) and PSS-TEMPO (red) in 1 M NaCl, at a scan rate of 50 mV s⁻¹ on a glassy carbon working electrode.

present the peak current vs. square root of scan rate ($v^{1/2}$) plots and the Ψ vs. negative square root of scan rate ($v^{-1/2}$) plots from the corresponding CVs for obtaining D and k_0 , respectively. Meanwhile, linear Levich plots (limiting current⁻¹ vs. $\omega^{-1/2}$) and Tafel plots from the corresponding LSV curves are shown in Figure S6e–h, where D and k_0 for (PPBPy)Br₂ and PSS-TEMPO can be determined based on the Koutecký-Levich equation. Table 2 compiles the electrochemical kinetics parameters of these two molecules. The D and k_0 values obtained from the two different methods are broadly consistent, in the order of magnitude of 10^{-6} cm² s⁻¹ and 10^{-3} cm s⁻¹ respectively for both (PPBPy)Br₂ and PSS-TEMPO, which are comparable with other reported organic and inorganic redox mediators.

Table 2: Electrochemical kinetics parameters of (PPBPy)Br₂ and PSS-TEMPO.

Redox molecule	Method	Diffusion coefficient [cm ² s ⁻¹]	Rate constant [cm s ⁻¹]
(PPBPy)Br ₂	Koutecký-Levich	3.91×10^{-6}	2.67×10^{-3}
	Nicholson	2.08×10^{-6}	3.38×10^{-3}
PSS-TEMPO	Koutecký-Levich	3.36×10^{-6}	5.29×10^{-3}
	Nicholson	5.08×10^{-6}	6.43×10^{-3}

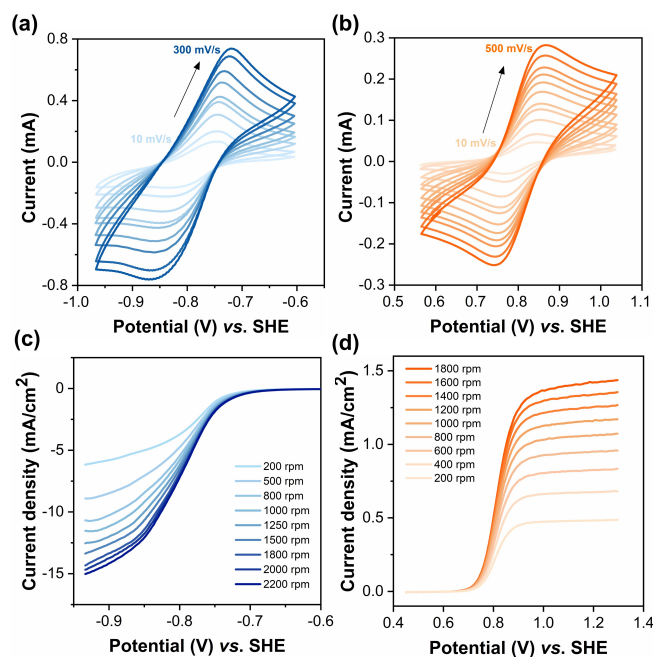


Figure 2. a, b) CV curves of (PPBPy)Br₂ and PSS-TEMPO at various scan rates for the calculation of D and k_0 by Nicholson method. c, d) LSV curves of (PPBPy)Br₂ and PSS-TEMPO at different rotating speeds for the calculation of D and k_0 by Koutecký-Levich equation. The concentration of (PPBPy)Br₂ was 50 mM in 2 M NaCl solution and that of PSS-TEMPO was 5 mM in 2 M NaCl solution. The scan rate of LSV in both tests was 10 mV s⁻¹.

To evaluate the species crossover rate, the permeabilities of 4-hydroxy TEMPO, (SPr)₂V, (PPBPy)Br₂ and PSS-TEMPO across an E620 K membrane were measured using a two-compartment H-cell. From Figure S7, the permeability of (PPBPy)Br₂ is determined to be around 1.17×10^{-12} cm² s⁻¹ through E620 K membrane, more than 100 times lower than that of (SPr)₂V. And that of PSS-TEMPO is around 5.36×10^{-12} cm² s⁻¹, more than 30 times lower than that of 4-hydroxy TEMPO. Thus, this (PPBPy)Br₂/PSS-TEMPO flow battery is expected to offer a reduced crossover compared with other AORFBs, which will enable a longer stand-by time with lowered self-discharge.

To investigate the chemical and electrochemical stability of (PPBPy)Br₂ and PSS-TEMPO, (PPBPy)Br₂/K₄Fe(CN)₆ and K₃Fe(CN)₆/PSS-TEMPO flow cells were assembled with E620 K membrane, respectively. Figure 3a shows the cycling performance of 0.10 M (PPBPy)Br₂/K₄Fe(CN)₆ cell. After 1000 cycles (>16 days), there wasn't noticeable capacity fading observed. We attribute the superior stability of (PPBPy)Br₂ to the strong electrostatic repulsion among molecules because of the negative charges and large steric hindrance of the phosphonate groups, which prevent the dimerization of the reduced species—the general degradation process of viologen derivatives.^[19] Meanwhile, the phenylene bridge, acting as an extended π -conjugated group, delocalizes the charges and further maintains its structural integrity upon reduction.^[16,20] Figure 3b shows the cycling of K₃Fe(CN)₆/PSS-TEMPO cell operated at a full charge–discharge mode for more than 20 days. The capacity faded

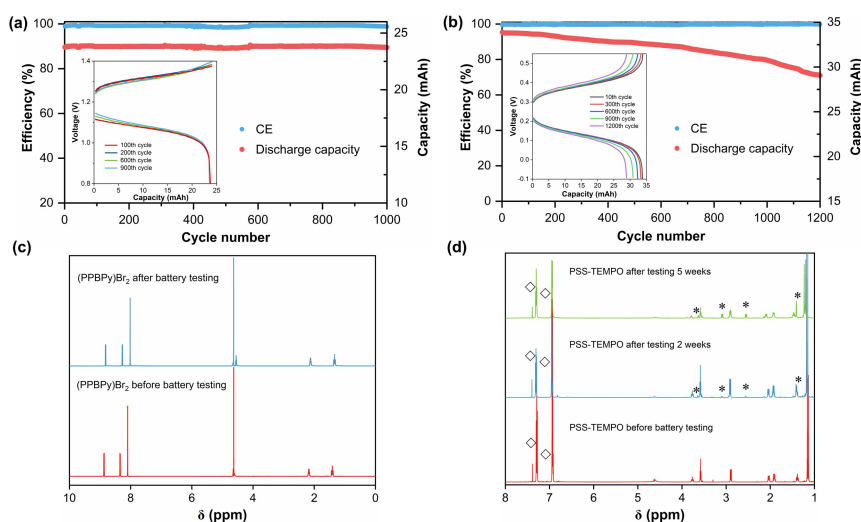


Figure 3. a) Cycling performance of 0.10 M (PPBPy)Br₂/0.20 M K₄Fe(CN)₆ flow cell at 30 mA cm⁻². The inset shows the voltage profiles of the cell at the 100th, 200th, 600th and 900th cycle, respectively. b) Cycling performance of 0.20 M K₃Fe(CN)₆/0.15 PSS-TEMPO flow cell at 40 mA cm⁻². The inset shows the voltage profiles of the cell at the 100th, 300th, 600th, 900th and 1200th cycle, respectively. c) ¹H NMR spectra of (PPBPy)Br₂ anolyte before and after battery cycling. An aliquot (0.5 mL) was collected after battery testing and D₂O (20% v/v) was added for the NMR signal locking purpose. The chemical shifts of all peaks were slightly shifted to up field due to the changes in pH of (PPBPy)Br₂ anolyte after battery testing (See overlap comparison in Figure S8). d) ¹H NMR spectra of PSS-TEMPO catholyte before and after (2 and 5 weeks) battery cycling. *: The peaks belong to decomposed by-product after cycling. ◇: The battery testing samples of PSS-TEMPO was reduced by phenyl hydrazine before NMR analysis and the peaks in the region from 7.30 to 6.90 ppm were assigned to phenyl hydrazine.

from 33.9 mAh to 29.0 mAh, corresponding to 0.012 %/cycle or 0.71 %/day, which is on par with the most stable TEMPO derivatives in literature.^[14]

To investigate the capacity decay mechanism during battery cycling processes, ¹H NMR and Electrospray Ionization Mass Spectrometry (ESI-MS) analyses were conducted by collecting the aliquots of the anolyte and catholyte before and after battery cycling. The ¹H NMR data in Figure 3c shows no perceptible decomposition observed for (PPBPy)Br₂ after cycling, confirming the outstanding stability of (PPBPy)Br₂. In comparison, the new peaks (marked as "*" in Figure 3d) that appear in the ¹H NMR of PSS-TEMPO indicate a degradation of the molecules after battery cycling. A close analysis of these peaks suggests that the changes should not be induced by bimolecular or multi-molecular annihilation, or ring-opening side reactions, but are likely due to the cleavage of nitroxyl radicals from N–O• to N–H. This was further confirmed by the mass spectrometric analysis (Figure S9). Note a similar degradation process has also been reported recently for a cationic TEMPO derivative by Pan *et al.*^[15] Here it is rational to infer that the degradation is associated with proton attack to the N–O•, for which excess protons could be generated by traces OER at the end of charging process, as substantiated by the slightly lowered pH of catholyte after cycling. To suppress the degradation, one potential approach is to employ high concentration supporting electrolyte (*i.e.*, ZnCl₂, NH₄Cl, *etc.*),^[12b] with which reportedly the abundant Cl⁻ ions could shield the oxoammonium cation and restrain the proton attack. Alternatively, replacing the methyl groups of TEMPO with bulkier ones is presumably another way of

protecting the nitroxyl radical, while it may also lead to reduced reaction kinetics.

Given the high chemical/electrochemical stability and low permeability of the redox pair, (PPBPy)Br₂/PSS-TEMPO full cells with different electrolyte concentrations were assembled with E620 K membrane, using the device shown in Figure S10. The cell capacity was set at full use of the one-electron reduction of (PPBPy)Br₂ while PSS-TEMPO was in excess. The low-concentration flow cell employed 0.15 M (PPBPy)Br₂ in anolyte and 0.20 M PSS-TEMPO in catholyte and was placed in a N₂-filled glovebox for extended galvanostatic cycling tests. As shown in Figure 4a, the battery operated at 40 mA cm⁻² for more than 600 cycles (11 days) without a capacity decay. The average coulombic efficiency (CE) was above 99.8 % and the energy efficiency (EE) was about 80 %. With the same cell, the rate performance was assessed subsequently. The cell was cycled at constant current densities from 10 to 100 mA cm⁻², after which the current was changed back to 40 mA cm⁻² with excellent retention of cell performance (Figure 4b). As shown in Figure 4c, another cell with 0.45 M (PPBPy)Br₂ and 0.60 M PSS-TEMPO was tested at 60 mA cm⁻² for more than 210 h, which demonstrated >99.7 % CE, ≈77.5 % EE, and similarly zero capacity fading in the 320-cycle test. In addition, stable cell performance was observed at current densities from 40 to 180 mA cm⁻² (Figure 4d). As the current was reduced to 50 mA cm⁻², nearly 100 % CE was attained with an EE of ≈80 %. Moreover, we have also tested a flow cell consisting of 0.45 M (PPBPy)Br₂/0.60 M PSS-TEMPO outside the glovebox, to reduce the ohmic resistance resulted from long cables and multiple connections (See Figure S11a). As shown in Figure 5a, at 100 mA cm⁻², the

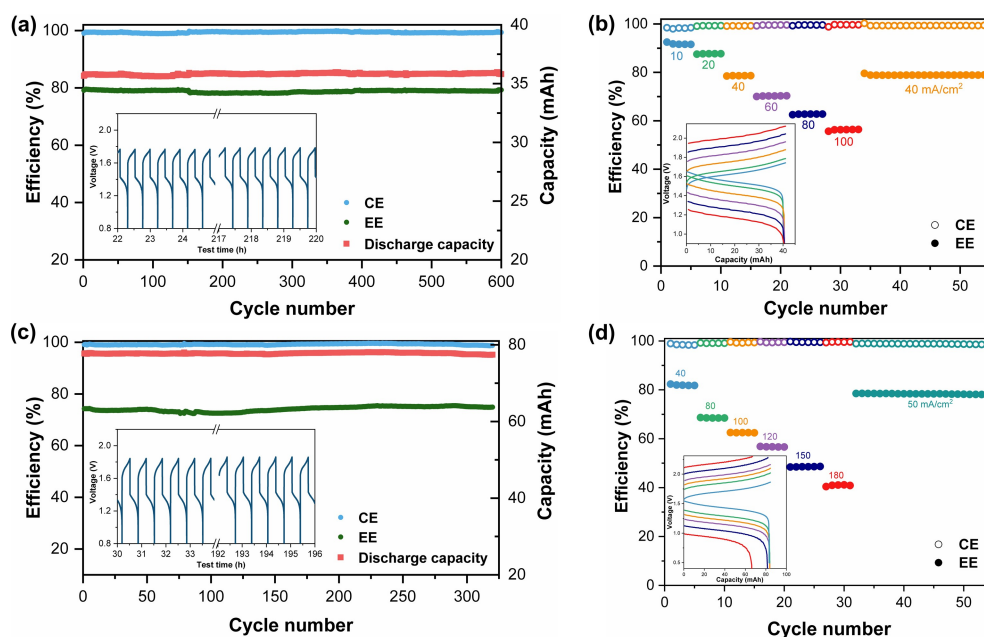


Figure 4. a) Cycling and b) rate performance of 0.15 M (PPBPy)Br₂/0.20 M PSS-TEMPO flow cell. c) Cycling and d) rate performance of 0.45 M (PPBPy)Br₂/0.60 M PSS-TEMPO flow cell. The current density was 40 mA cm⁻² in (a) and 60 mA cm⁻² in (c). The insets show the voltage profiles of the two cells in respective galvanostatic test. All the batteries were tested in a N₂-filled glove box.

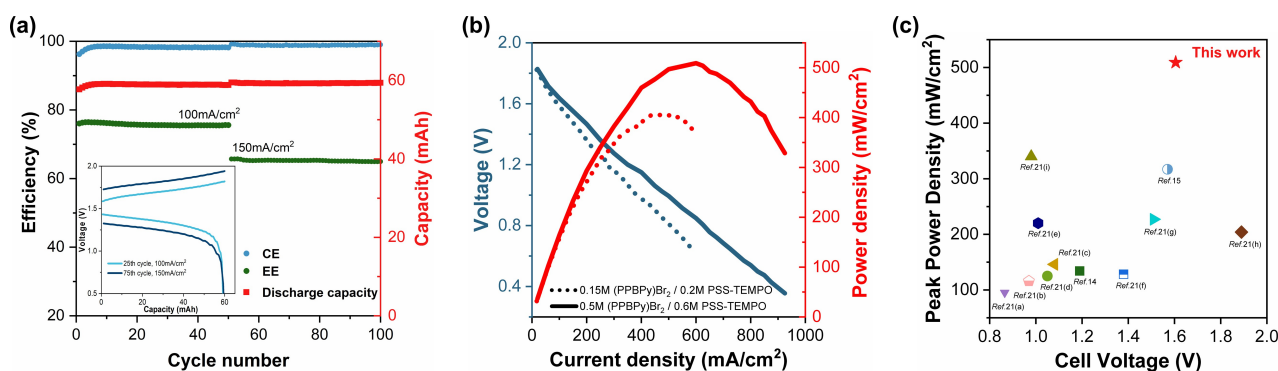


Figure 5. a) A (PPBPy)Br₂/PSS-TEMPO full cell tested with reduced series resistance. Anolyte: 5 mL of 0.45 M (PPBPy)Br₂ in 2 M NaCl. Catholyte: 10 mL of 0.6 M PSS-TEMPO in 2 M NaCl. b) Polarization curves and power densities of the two (PPBPy)Br₂/PSS-TEMPO flow cells at 100 % SOC. c) Comparisons of OCV and peak power density between this work and other reported pH-neutral AORFBs. See Table S1 for detailed information.

EE was boosted to 76%. When the current density was further increased to 150 mA cm⁻², the discharge voltage plateau still maintained at 1.25–1.30 V, which is comparable to that of VFBs. We believe with the same battery chemistry, there is ample room to further improve the power output by optimizing the cell components.

With the high cell voltage, fast electron transfer kinetics and the use of low resistance CEM, this (PPBPy)Br₂/PSS-TEMPO full cell is poised to offer superior power performance. Polarization curves of both the 0.15 M (PPBPy)Br₂/0.20 M PSS-TEMPO and 0.50 M (PPBPy)Br₂/0.60 M PSS-TEMPO cells were measured at 100 % SOC (Figure 5b). For the low-concentration cell, a peak power density of 405 mW cm⁻² was attained, while a maximum power density of 509 mW cm⁻² was achieved with the high-concentration cell at a current density of ≈ 600 mA cm⁻², which is the

highest value among all reported pH-neutral RFBs (Figure 5c) regardless of organic or inorganic redox species.^[14,15,21] Note in the above tests, we didn't go to higher concentrations primarily due to an increased viscosity of the anolyte, presumably induced by the reduced (PPBPy)Br₂. Note that the area-specific resistance of E620(K) membrane was 1.05 Ω cm² in 2 M NaCl solution as determined by impedance spectroscopy measurement, whereas that of Nafion 212 was 0.77 Ω cm² (Figure S11b). Thus, the (PPBPy)Br₂/PSS-TEMPO cell would potentially present an even higher power performance using Nafion 212. However, based on the previous studies,^[10] the selectivity of E620(K) is much higher than Nafion 212. E620(K) was thus employed as the separator to demonstrate the battery performance in this study.

Near-neutral AORFBs with varying cycling stability have been extensively reported,^[22] from which there is seemingly a trade-off between cell voltage and lifetime, as governed by the thermodynamic stability of redox species and side reactions of other species in the electrolytes. By examining the OCV vs. lifetime among the reported ferrocene and TEMPO-based cells, the MV/FcNCl cell (1.06 V) has a capacity fading rate of 1.3 % per day,^[21d] while the BTMAP-Vi/BTMAP-Fc pair (0.75 V) with a lower OCV shows greatly enhanced stability with a capacity fading rate of 0.1 % per day.^[17] Similarly, the BTMAP-Vi/TMAP-TEMPO (1.19 V) cell is reported to be more robust than MV/TEMPTMA (1.42 V) and [(NPr)₂TTz]Cl₄/NMe-TEMPO (1.44 V).^[13,14,20] Interestingly, despite the anionic (PPBPy)Br₂/PSS-TEMPO pair (1.61 V) demonstrated here having a much higher cell voltage than its cationic counterpart—the BTMAP-Vi/TMAP-TEMPO cell, the capacity fading (0.71 %/day at 100 % SOC in terms of PSS-TEMPO) is comparable. Nevertheless, as the cell voltage is beyond the stability window of water, while it could to some extent be suppressed by using suitable electrode materials, the side reactions (*i.e.*, HER or OER) may always occur during the charging process, as corroborated by the pH changes of both anolyte and catholyte. It was observed that, with the self-limited HER and OER processes, the pH of anolyte and catholyte which were initially neutral stabilized at around 10.5 and 4.5 respectively after prolonged operation. Alternatively, as shown in Figure 3a, given the lability of PSS-TEMPO, the cell stability could be addressed by switching the catholyte to [Fe(CN)₆]^{3-/4-}, with which an ultra-stable battery system with a cell voltage of 1.27 V could be obtained.

Conclusion

In summary, two anionic redox-active molecules, (PPBPy)Br₂ and PSS-TEMPO which are compatible with the commonly used CEMs at near pH-neutral conditions are designed and prepared with feasible synthetic routes for high-performance AORFB applications. These two molecules have a potential gap as high as 1.61 V and exhibited remarkably low permeabilities ($\approx 10^{-12}$ cm² s⁻¹) through CEM. As an anolyte redox species, (PPBPy)Br₂ presented excellent cycling performance for more than 1000 cycles when paired with [Fe(CN)₆]^{3-/4-} in the catholyte. PSS-TEMPO, with a solubility higher than 2.55 M, showed a capacity fading rate about 0.7 % per day when used as a catholyte redox species. The degradation mechanism of PSS-TEMPO was deciphered to be ascribed to bond cleavage of the nitroxyl group upon proton attack, which was associated with the self-limited OER reaction. The (PPBPy)Br₂/PSS-TEMPO full cell has demonstrated good cycling stability with excess catholyte and a record power density of 509 mW cm⁻² among all reported near-neutral aqueous RFBs. We anticipate the molecular engineering strategy demonstrated here would provide useful means for the development of robust redox active species compatible

with CEMs for low-cost, high-safety and durability, and high-performance AORFBs.

Supporting Information

Essential experimental procedures, supplementary data, supplementary figures and tables can be found in Supporting Information.

Author Contributions

Q.W. and M.G. conceived the study. M.G. performed most of the experiments. M.S. synthesized (PPBPy)Br₂ and PSS-TEMPO and conducted the NMR tests. Q.W. supervised the work. Y.S. assisted with battery tests and figures preparation. M.G. and Q.W. wrote the paper.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: All-Organic Redox Flow Battery • Aqueous Organic Redox Flow Battery • Molecular Engineering • pH-Neutral

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